

Corner points, cusps, points with vertical/horizontal tangent

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Right derivative and left derivative



Introduction: Right derivative and left derivative

- ▶ In this section we introduce the **right derivative** and the **left derivative** of a function at a point.
- ▶ They are obtained by considering the right and left limits of the difference quotient.
- ▶ They are the tool to classify the points where a continuous function is **not differentiable**: corner points, cusps, points with vertical tangent.

Right and left derivative

Definition 1: of right (left) derivative

Let $f : (a, b) \rightarrow \mathbb{R}$; the function f is said to be differentiable at $x_0 \in (a, b)$ from the right (from the left) if the following limit exists and is finite

$$\lim_{h \rightarrow 0^+} \frac{f(x_0 + h) - f(x_0)}{h} \quad \left(\lim_{h \rightarrow 0^-} \frac{f(x_0 + h) - f(x_0)}{h} \right)$$

then f is differentiable from the right (from the left) and the limit is called the **right derivative** (**left derivative**).

- ▶ The right derivative is denoted by the symbol $f'_+(x_0)$, while the left derivative is denoted by the symbol $f'_-(x_0)$.

Summary

Right and left derivative:

- ▶ $f'_+(x_0)$: finite limit of the difference quotient as $h \rightarrow 0^+$
- ▶ $f'_-(x_0)$: finite limit of the difference quotient as $h \rightarrow 0^-$

Corner points

Introduction: Corner points

- ▶ In this section we define **corner points**: points where the function is continuous, has right and left derivatives, but is not differentiable.
- ▶ The typical example is the absolute value function $|x|$ at the origin.
- ▶ We will also derive the formula for the derivative of $|x|$ away from the origin.

Definition 2: of corner point

*If f is continuous and differentiable from the right and from the left (but not differentiable) at x_0 , we say that f has a **corner point** at $x = x_0$.*

Example: corner points (1/3)

Consider the function $f(x) = |x|$ and the point $x_0 = 0$. We have:

$$f'_+(0) = \lim_{h \rightarrow 0^+} \frac{|h|}{h} = \lim_{h \rightarrow 0^+} \frac{h}{h} = 1$$

and

$$f'_-(0) = \lim_{h \rightarrow 0^-} \frac{|h|}{h} = \lim_{h \rightarrow 0^-} \frac{-h}{h} = -1$$

Since the limit of the difference quotient does not exist, f is not differentiable at $x = 0$.

Example: corner points (2/3)

The function is continuous at $x = 0$ (the origin) since

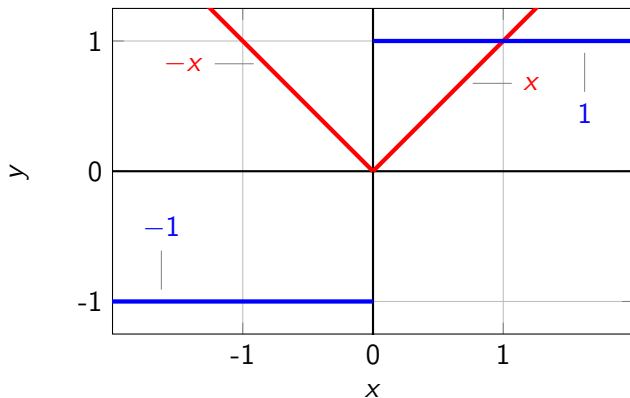
$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} x = 0 \quad \text{and} \quad \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} -x = 0$$

hence

$$\lim_{x \rightarrow 0} f(x) = f(0) = 0$$

Since the right and left limits of the difference quotient at $x = 0$ exist and are finite, the graph therefore has a corner point at $x = 0$.

Example: corner points (3/3)



Derivative of the absolute value

The formula that concisely expresses the derivative of the absolute value function (away from the origin) is the following:

$$f(x) = |x|, \quad f'(x) = \operatorname{sgn}(x) = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \end{cases}$$

Summary

Corner points:

- ▶ f continuous at x_0 , with $f'_+(x_0)$ and $f'_-(x_0)$ finite but different: corner point
- ▶ Example: $|x|$ at $x = 0$, with $f'_+(0) = 1$ and $f'_-(0) = -1$
- ▶ Away from the origin: $\frac{d}{dx}|x| = \operatorname{sgn}(x)$

Points with vertical/horizontal tangent



Introduction: Points with vertical/horizontal tangent

- ▶ In this section we study what happens when the limit of the difference quotient is **infinite** or is **zero**.
- ▶ In the first case the function is not differentiable, but the graph has a **vertical** tangent; in the second case the tangent is **horizontal**.
- ▶ We will see examples with cube roots.

Vertical tangent

- ▶ If f is continuous at a point x_0 and

$$\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} = \pm\infty$$

then f is not differentiable at x_0 but, geometrically, the graph of f has a well-defined tangent line parallel to the y -axis.

- ▶ In this case we will allow the notation

$$f'(x_0) = +\infty, \quad f'(x_0) = -\infty$$

and we will speak of a **point with vertical tangent**.

Point with vertical tangent

Definition 3: of point with vertical tangent

If

$$f'(x_0) = +\infty \quad \text{or} \quad f'(x_0) = -\infty$$

*we say that f has a **point with vertical tangent** at $x = x_0$.*

Horizontal tangent

- Similarly, if f is continuous at a point x_0 and

$$\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} = 0$$

the graph of f has a well-defined tangent line parallel to the x -axis. In this case we will speak of a **point with horizontal tangent**.

Definition 4: of point with horizontal tangent

If

$$f'(x_0) = 0$$

we say that f has a **point with horizontal tangent** at $x = x_0$.

Example: point with vertical tangent (1/3)

Let

$$f(x) = \sqrt[3]{x}$$

then for $x_0 = 0$

$$\frac{f(h) - f(0)}{h} = \frac{\sqrt[3]{h}}{h} = \frac{1}{h^{2/3}}$$

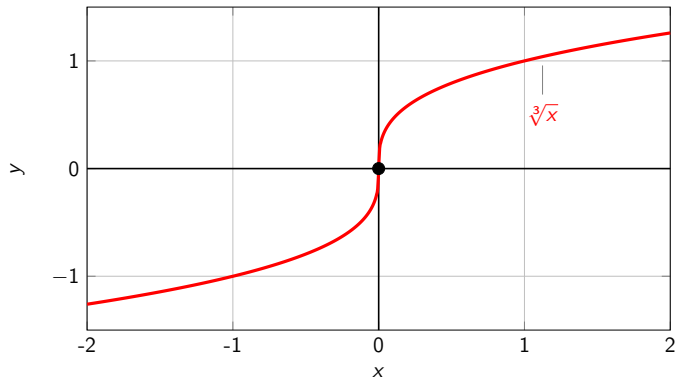
and the limits are:

$$\lim_{h \rightarrow 0^-} \frac{1}{h^{2/3}} = +\infty \quad \text{and} \quad \lim_{h \rightarrow 0^+} \frac{1}{h^{2/3}} = +\infty$$

hence

$$\lim_{h \rightarrow 0} \frac{1}{h^{2/3}} = +\infty \quad \text{and} \quad f'(0) = +\infty.$$

Example: point with vertical tangent (2/3)



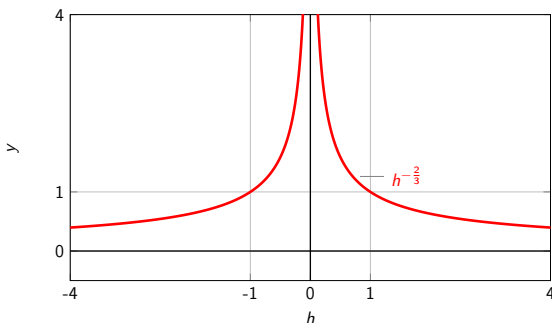
The function has a point with vertical tangent at $x_0 = 0$.

Example: point with vertical tangent (3/3)

The function:

$$f(h) = \frac{1}{h^{2/3}} = h^{-\frac{2}{3}}$$

is a power with negative rational exponent $\frac{m}{n}$ with n odd (hence defined on $\mathbb{R} \setminus \{0\}$) and m even (hence an even function). Its graph is:



Example: point with vertical tangent (1/2)

Let

$$f(x) = -\sqrt[3]{x-2} + 1$$

then for $x_0 = 2$

$$\frac{f(2+h) - f(2)}{h} = \frac{-\sqrt[3]{(2+h)-2} + 1 - 1}{h} = -\frac{\sqrt[3]{h}}{h} = -\frac{1}{h^{2/3}}$$

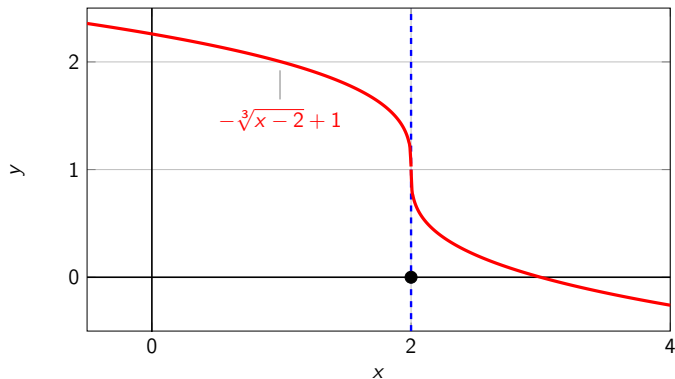
and the limits are

$$\lim_{h \rightarrow 0^-} -\frac{1}{h^{2/3}} = -\infty \quad \text{and} \quad \lim_{h \rightarrow 0^+} -\frac{1}{h^{2/3}} = -\infty$$

hence

$$\lim_{h \rightarrow 0} -\frac{1}{h^{2/3}} = -\infty \quad \text{and} \quad f'(2) = -\infty.$$

Example: point with vertical tangent (2/2)



The function has a point with vertical tangent at $x_0 = 2$.

Summary

Points with vertical/horizontal tangent:

- ▶ f continuous at x_0 and $f'(x_0) = \pm\infty$: point with **vertical tangent** (f not differentiable)
- ▶ $f'(x_0) = 0$: point with **horizontal tangent**
- ▶ Examples: $\sqrt[3]{x}$ at $x_0 = 0$ ($f'(0) = +\infty$), $-\sqrt[3]{x-2} + 1$ at $x_0 = 2$ ($f'(2) = -\infty$)

Cusps

Introduction: Cusps

- ▶ In this section we define **cusps**: points where the right derivative and the left derivative are infinite with opposite signs.
- ▶ We will also discuss the mixed cases and the boundary points of the domain with infinite derivative.
- ▶ We will conclude with some examples of functions **extended by continuity** from the right at the origin.

Definition 5: of cusp

Let f be a function continuous at x_0 ; if

$$f'_+(x_0) = \pm\infty \quad \text{and} \quad f'_-(x_0) = \mp\infty$$

we say that f has a cusp at x_0 .

Example: cusp (1/2)

Let

$$f(x) = \sqrt[3]{|x|}$$

then for $x_0 = 0$

$$\frac{f(h) - f(0)}{h} = \frac{\sqrt[3]{|h|}}{h}$$

and the limits are:

$$\lim_{h \rightarrow 0^-} \frac{\sqrt[3]{-h}}{h} = \lim_{h \rightarrow 0^-} -\frac{1}{h^{2/3}} = -\infty$$

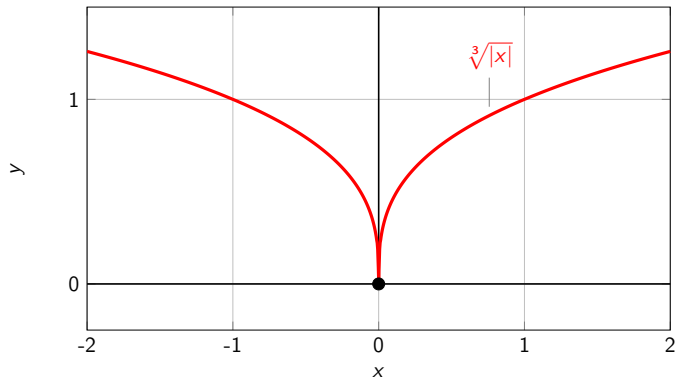
and

$$\lim_{h \rightarrow 0^+} \frac{\sqrt[3]{h}}{h} = \lim_{h \rightarrow 0^+} \frac{1}{h^{2/3}} = +\infty,$$

hence:

$$f'_-(0) = -\infty \quad \text{and} \quad f'_+(0) = +\infty.$$

Example: cusp (2/2)



The function has a cusp at $x_0 = 0$.

Mixed cases and boundary points

- ▶ In the mixed case where one of the two derivatives is finite and the other is infinite (with f continuous) we still speak of a corner point.
- ▶ Finally, if the function is defined only for $x \geq x_0$ and at that point has an infinite (right) derivative, we will simply say that at that point it has a vertical tangent, without speaking of either a cusp or a point with vertical tangent.

Example: point with vertical tangent (1/3)

Let

$$f(x) = \sqrt{x}$$

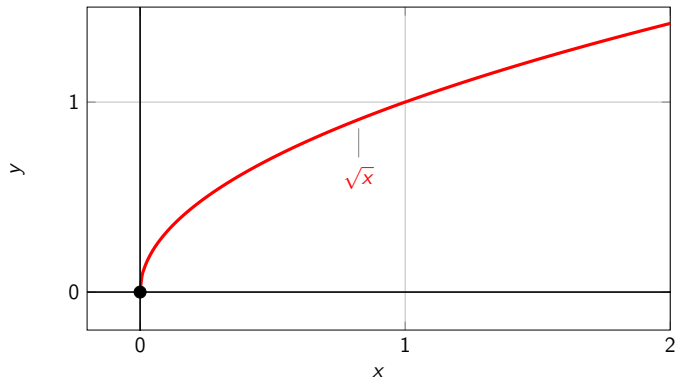
then for $x_0 = 0$

$$\frac{f(h) - f(0)}{h} = \frac{\sqrt{h}}{h} = \frac{1}{h^{1/2}}$$

we only have the right limit

$$\lim_{h \rightarrow 0^+} \frac{1}{h^{1/2}} = +\infty \quad \text{and} \quad f'_+(0) = +\infty.$$

Example: point with vertical tangent (2/3)



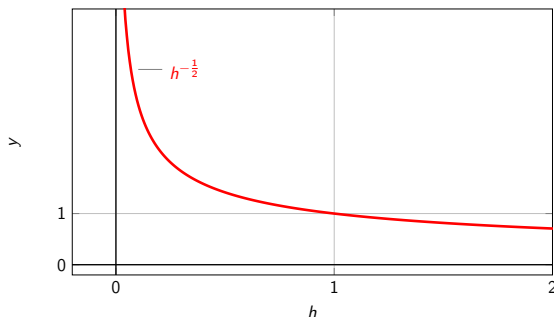
The function has a point with vertical tangent at $x_0 = 0$.

Example: point with vertical tangent (3/3)

The function:

$$f(h) = \frac{1}{h^{1/2}} = h^{-\frac{1}{2}}$$

is a power with negative rational exponent $\frac{m}{n}$ with n even (hence defined only on $\mathbb{R}_+$). Its graph is:



Example: continuous extension from the right (1/3)

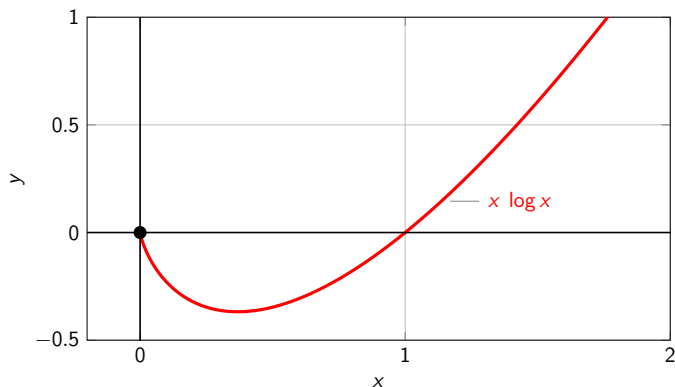
Let

$$f(x) = x \log x, \quad \text{for } x > 0 \quad \text{we have} \quad \lim_{x \rightarrow 0^+} x \log x = 0$$

hence the function can be extended by continuity from the right at $x = 0$, by setting:

$$f(x) = \begin{cases} x \log x & \text{if } x > 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Example: continuous extension from the right (2/3)



Example: continuous extension from the right (3/3)

We compute the right derivative at $x_0 = 0$ of $f(x) = x \log x$ extended by continuity from the right at $x = 0$. We have, for $x_0 = 0$:

$$\frac{f(x_0 + h) - \overbrace{f(x_0)}^{=0}}{h} = \frac{f(x_0 + h)}{h} = \frac{f(h)}{h}$$

and consequently

$$\lim_{h \rightarrow 0^+} \frac{f(x_0 + h) - f(x_0)}{h} = \lim_{h \rightarrow 0^+} \frac{h \log h}{h} = \lim_{h \rightarrow 0^+} \log h = -\infty$$

hence

$$f'_+(x_0) = -\infty$$

The function therefore has a point with vertical tangent at $x_0 = 0$.

Example: continuous extension from the right (1/3)

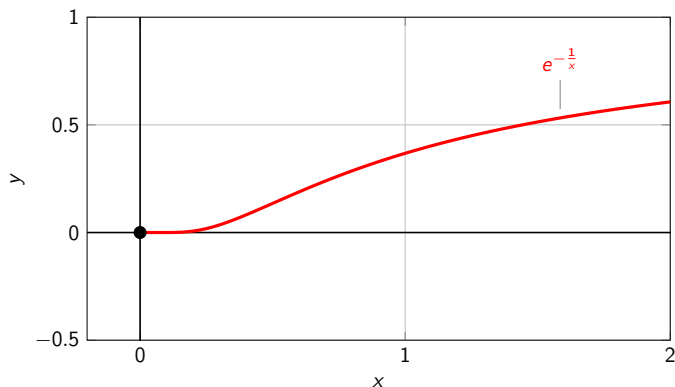
Let

$$f(x) = e^{-\frac{1}{x}}, \text{ for } x > 0 \text{ we have } \lim_{x \rightarrow 0^+} e^{-\frac{1}{x}} = 0$$

hence the function can be extended by continuity from the right at $x = 0$, by setting:

$$f(x) = \begin{cases} e^{-\frac{1}{x}} & \text{if } x > 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Example: continuous extension from the right (2/3)



Example: continuous extension from the right (3/3)

We compute the right derivative at $x_0 = 0$ of $f(x) = e^{-\frac{1}{x}}$ extended by continuity from the right at $x = 0$. We have:

$$\lim_{h \rightarrow 0^+} \frac{f(x_0 + h) - f(x_0)}{h} = \lim_{h \rightarrow 0^+} \frac{e^{-\frac{1}{h}}}{h} = \lim_{h \rightarrow 0^+} \frac{1}{e^{\frac{1}{h}} h}$$

Changing the variable:

$$y = \frac{1}{h}, \quad \text{se } h \rightarrow 0^+ \quad \text{then } y \rightarrow +\infty$$

We obtain:

$$\lim_{h \rightarrow 0^+} \frac{1}{e^{\frac{1}{h}} h} = \lim_{y \rightarrow +\infty} \frac{y}{e^y} = 0 \quad \text{hence} \quad f'_+(x_0) = 0$$

and the function has a point with horizontal tangent (from the right) at $x_0 = 0$.

Summary

Cusps:

- ▶ f continuous at x_0 with $f'_+(x_0) = \pm\infty$ and $f'_-(x_0) = \mp\infty$: cusp (e.g. $\sqrt[3]{|x|}$ at 0)
- ▶ One derivative finite and the other infinite: still a corner point
- ▶ f defined only for $x \geq x_0$ with infinite right derivative: vertical tangent (e.g. $\sqrt{x}$ at 0)
- ▶ Extensions by continuity: $x \log x$ has $f'_+(0) = -\infty$, $e^{-\frac{1}{x}}$ has $f'_+(0) = 0$

Key messages

Summary

Classification of the points of non-differentiability of a continuous function:

- ▶ **Corner point:** $f'_+(x_0) \neq f'_-(x_0)$, at least one of the two finite
- ▶ **Point with vertical tangent:** $f'(x_0) = +\infty$ or $f'(x_0) = -\infty$
- ▶ **Cusp:** $f'_+(x_0) = \pm\infty$ and $f'_-(x_0) = \mp\infty$

Moreover:

- ▶ **Point with horizontal tangent:** $f'(x_0) = 0$